

A Stochastic Recurrence with Sum and Max

1.1 Definition

Consider the recurrence

$$Y_{n+2} \stackrel{d}{=} I^{(n)} \left(Y_{U_n}^{(1)} + Y_{n+1-U_n}^{(2)} + 1 \right) + J^{(n)} \max \left(Y_{U_n}^{(1)}, Y_{n+1-U_n}^{(2)} \right), \quad Y_1 = Y_2 = 0. \quad (1.1)$$

The context leading to analyzing this recurrence is given in Section 1.4. The variables I and J are Bernoulli with expectation u and v respectively, such that $I + J \leq 1$. The variable U_n is uniformly distributed on $\{1, \dots, n\}$. Superscripts are for independent copies. We assume that all collections of copies are independent. We want to analyze the asymptotic of Y_n .

1.2 Lower Bounds

Taking the expectation and using $\max(Y_i, Y_{n+1-i}) \geq \frac{1}{2}(Y_i + Y_{n+1-i})$, we get a lower bound sequence

$$y_{n+2} = u + \frac{2u+v}{n} \sum_{i=1}^n y_i, \quad y_1 = y_2 = 0. \quad (1.2)$$

In the case $2u+v > 1$ which aligns with real data (Section 1.4), we have $y_n \sim C' n^{2u+v-1}$. Hence we want to prove or disprove that in such case, $\mathbb{E}[Y_n] \sim C n^\alpha$ for some $\alpha > 2u+v-1$.

Another lower bound is obtained by using

$$\mathbb{E}[\max(Y_i, Y_{n+1-i})] \geq \max(\mathbb{E}[Y_i], \mathbb{E}[Y_{n+1-i}]) = \max(y_i, y_{n+1-i}) = y_{\max(i, n+1-i)}.$$

That gives another (homogenized) lower bound sequence

$$x_{n+2} = \frac{2u}{n} \sum_{i=1}^n x_i + \frac{2v}{n} \sum_{i=\lfloor (n+1)/2 \rfloor}^n x_i, \quad n \geq 2, \quad x_1 = x_2 = 1. \quad (1.3)$$

It is proved by induction that $y_n \in \Omega(n^\beta)$, where β is the unique solution to

$$2(u+v) - v2^{-\beta} = \beta + 1.$$

1.3 Attempts

We want to see if there is α and $C > 0$ such that $X_n = \frac{Y_n}{C n^\alpha}$ converges in distribution to a random variable X of mean 1. Substituting in the recurrence, we get the limit distributional equation

$$X \stackrel{d}{=} I \left(U^\alpha X^{(1)} + (1-U)^\alpha X^{(2)} \right) + J \max \left(U^\alpha X^{(1)}, (1-U)^\alpha X^{(2)} \right), \quad (1.4)$$

where U is uniformly distributed on $(0,1)$. A subproblem is to prove or disprove that *there exists a unique α such that the equation has a solution with mean 1*. Contraction method with Wasserstein or Zolotarev metrics did not work. That may be because we need to exploit more information of α to prove contraction in a more appropriate space.

1.4 Context of the Recurrence

Note. This section can be skipped.

Hyperscan is a regex matching engine. It works based on decomposing regexes into smaller literals and FAs. According to real data [1], a pattern consists of many literals. This is confirmed by fitting the dataset to a BST-like model [2] using maximum likelihood. Details are as follows.

Let $\mathcal{A}$ be a finite alphabet and $\mathcal{T}$ be the set of all regexes excluding ϵ over $\mathcal{A}$.

Definition 1 (BST-like distribution for regexes). *We define $p : \mathcal{T} \rightarrow [0, 1]$ as follows.*

1. $\sum_{c \in \mathcal{A}} p(c) = 1,$
2. $p\left(\begin{array}{c} \bullet \\ / \backslash \\ T_1 \quad T_2 \end{array}\right) = \frac{u}{n-2} p(T_1) p(T_2), \text{ where } |T_1| + |T_2| = n - 1.$
3. $p\left(\begin{array}{c} + \\ / \backslash \\ T_1 \quad T_2 \end{array}\right) = \frac{v}{n-2} p(T_1) p(T_2), \text{ where } |T_1| + |T_2| = n - 1.$
4. $p\left(\begin{array}{c} * \\ | \\ T \end{array}\right) = (1 - u - v) p(T).$

We want to compute the expected size of the largest all-concatenation component in a regex over all of equivalent forms of its parse tree. The reason is as follows. Take the regex

$$(abc + d)(xy + z) = abcx y + abcx z + dxy + dz.$$

We expect the largest concatenation-component to be in $abcx y$ instead of abc because the former is extracted by Hyperscan. Therefore, a subtree may *yield* its concatenation-component to its parent if the parent is a concatenation node. Recall the construction

$$\mathcal{T} = \sum_{c \in \mathcal{A}} c + \begin{array}{c} * \\ | \\ \mathcal{T} \end{array} + \begin{array}{c} \bullet \\ / \backslash \\ \mathcal{T} \quad \mathcal{T} \end{array} + \begin{array}{c} + \\ / \backslash \\ \mathcal{T} \quad \mathcal{T} \end{array}.$$

Define the random variables

1. M_n : the largest concatenation-component that is not inside a Kleene star of a regex of size n *itself*,
2. Y_n : the largest concatenation-component that is not inside a Kleene star of a regex of size n that it *yields* to its parent.

We have the following recurrences

$$Y_{n+2} = \begin{cases} Y_{U_n}^{(1)} + Y_{n+1-U_n}^{(2)} + 1 & \text{prob. } u, \\ \max(Y_{U_n}^{(1)}, Y_{n+1-U_n}^{(2)}) & \text{prob. } v, \\ 0 & \text{prob. } 1 - u - v \end{cases} \quad Y_1 = Y_2 = 0. \quad (1.5)$$

$$M_{n+2} = \begin{cases} \max(Y_{U_n}^{(1)} + Y_{n+1-U_n}^{(2)} + 1, M_{U_n}^{(1)}, M_{n+1-U_n}^{(2)}) & \text{prob. } u, \\ \max(M_{U_n}^{(1)}, M_{n+1-U_n}^{(2)}) & \text{prob. } v \\ 0 & \text{prob. } 1 - u - v \end{cases} \quad M_1 = M_2 = 0. \quad (1.6)$$

We have used $Y_n^{(1)}, Y_n^{(2)}, M_n^{(1)}, M_n^{(2)}$ as independent copies of Y_n and M_n respectively, and U_n as a uniform random variable over $\{1, 2, \dots, n\}$ independent of all $Y_n^{(i)}$ and $M_n^{(i)}$. The recurrence

(1.1) is the distributional version of (1.5). Note that we can reuse the same U_n and $Y_n^{(i)}$ since the branches to which they are applied cannot be both active.

Due to MLE estimation, it is plausible to assume that $2u + v > 1$ and even $u > v$.

References

- [1] Cisco Systems. *Snort Rules Database*. <https://www.snort.org/>. Accessed: 2026-05-18. 2026.
- [2] Cyril Nicaud, Carine Pivoteau, and Benoît Razet. “Average analysis of Glushkov automata under a BST-like model”. In: *30th Foundations of Software Technology and Theoretical Computer Science (FSTTCS’10)*. Vol. 8. Schloss Dagstuhl. 2010, pp. 388–399.